

EEE4114F: ML Tutorial 1

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Question 1

Suppose you have been asked to develop a system that takes images of a tube containing two-phase flows (e.g. a mixture of a vapour and liquid. See Figure 1 for an illustration) in order to determine the type of flow (bubbly, slug etc.) as well as predict a quantity known as vapour quality (how much of a mixture is a vapour).

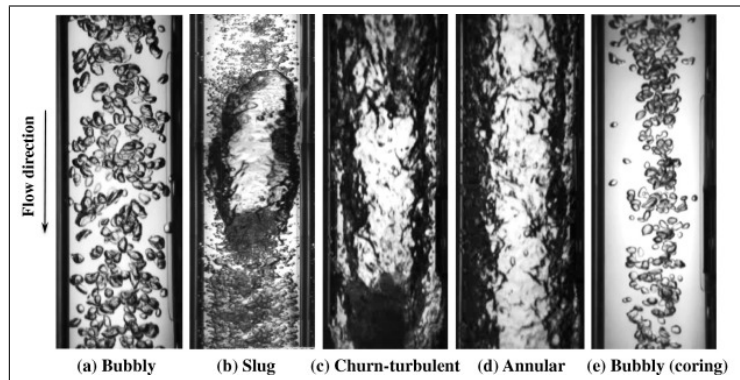


Figure 1: An example of images that indicated different types of two-phase flow.

The input data consists of greyscale images that have a resolution of $320\text{ px} \times 640\text{ px}$. The target data includes text for each class as well as the value of the vapour quality measured during the capturing of each image e.g. $y = (\text{"bubbly"}, 0.7)$. Answer the following questions:

- Is this a supervised learning/unsupervised learning/reinforcement learning scenario?
- How many input features would be used by your models?
- What types of learning task need to be solved? What aspects of the problem relate to each task? What component of the target data would you use for each task?
- The differences between the different flows are not well defined and scientists have asked you to find different ways of grouping the flows based only on the images. What kind of learning scenario is this?
- You train a model that produces the training and validation loss curves as shown in Figure 2. For each of the marked points (A/B/C/D) during the training process describe how you would expect the model to perform, and indicate which would be the point at which the model parameters would likely result in the best performance in the real world.

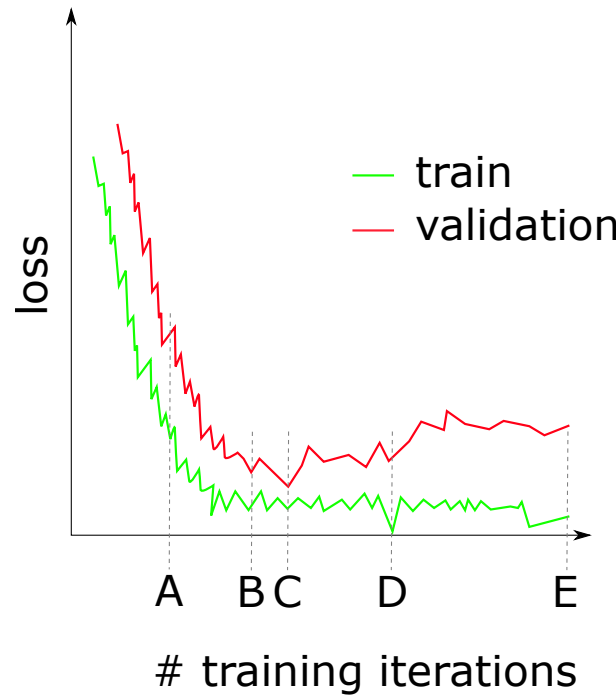


Figure 2: Loss curves during the training of a model

Question 2

Suppose you have a new low-cost, analogue pressure sensor, but there is no datasheet available and you do not know how the sensor's output voltage relates to pressure. You decide to try and characterise it by measuring the output voltage from the low-cost sensor with a multimeter along with pressure readings from a known sensor.

You collect the following data where each tuple represents an input-output pair (x_i, y_i) .

$$\mathcal{D} = \{(15, 5.2), (20, 6.3), (25, 7.8)\}$$

You then decide to fit a linear regression model of the form $h_{\theta}(x) = \theta_0 + \theta_1 x$ to it using gradient descent optimization. The model's parameters are randomly initialised such that $\theta_0 = 0.4, \theta_1 = 0.2$. The objective function, J , you choose to optimize is given by the mean square error:

$$J = \frac{1}{2N} \sum_{i=1}^N (h_{\theta}(x_i) - y_i)^2$$

Where N is the number of examples in a batch. Answer the following questions:

- For this scenario what is the input feature and what is the output variable?
- Calculate the initial loss, J , for the full batch of data.
- Write down the expressions for the gradients of the objective function with respect to the model's parameters.
- Calculate the gradient for θ_0
- Calculate the gradient for θ_1
- If the learning rate is, $\eta = 0.001$, calculate the new model parameters (round to three decimal places) after a single update.
- You find that you model takes too long to train so you increase the learning rate to $\eta = 0.1$. What is the danger of increasing the learning rate too much?
- You start to wonder if there are other phenomena might impact this sensor e.g. perhaps temperature influences the sensor readings.

- (i) How could you modify the model to incorporate one additional feature? Write down the new form for $h_{\theta}(x)$.
 - (ii) How would adding this additional feature impact the variance of the model?
- i. You decide you need more data and collect more of it to increase the the number of training examples. You train your model again until the training loss is low enough. You use this trained model in your final system that outputs a pressure value based on the sensor reading.
 - (i) Your model does not seem to output accurate values when you implement it in the real world. Why is it not performing as you expected?
 - (ii) You perform a three-way holdout sampling technique multiple times, but your validation estimates are a poor estimate of the test performance. What resampling technique could you use that would improve this? Explain how it works and why it should produce better estimates.
- j. If it turns out that both features have a non-negligible impact on the sensor's output would it be more appropriate to use l_1 or l_2 penalties to regularize the objective function? Explain your answer.

END